Physics 7B

UC Berkeley

Spring 2026

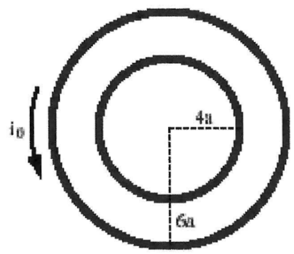
7B Discussion Problems, Week 12

Topics
Ampere's Law and the Biot-Savart Law

Week 12 Discussion Problems

Ampere's Law and the Biot-Savart Law

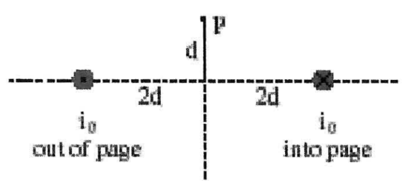
1. Two circular wire loops are fixed in place as shown. The large loop carries current i_0 counter-clockwise as viewed from above.



   (a) Suppose we want the magnetic field to vanish at the common center of the loops. Should we generate a clockwise current in the small loop or a counter-clockwise current?

   (b) How strong a current will we need to generate? (Hint: At the center of a circular current loop of radius R the loops magnetic field has magnitude $B_{\text{loop}} = \frac{\mu_0 I}{2R}$.)

2. Two straight wires are fixed in place near one another, and carry equal currents i_0 in opposite directions.

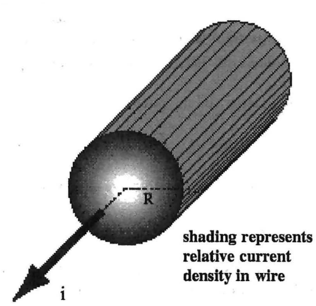


   (a) Sketch the two contributions to the net magnetic field at point P.

   (b) Add these vectors graphically to obtain a sketch of the net magnetic field vector at P.

   (c) Calculate the magnitude and direction of the magnetic field at P.

3. A thick, infinite wire of radius R carries total current i distributed *non-uniformly* across its cross section. The current density within the wire (in amps per square meter) is given by $j(r) = j_0(r/R)^4$.



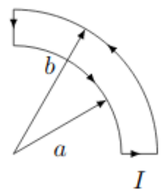
   (a) Sketch the magnetic field created by the wire, at points both inside and outside the wire.

   (b) Find the strength of the magnetic field created by the wire, at any point inside the wire ($r < R$).

   (c) Find the strength of the magnetic field created by the wire, at any point outside the wire ($r > R$).

4. Problem:

Problem 20.4: (Magnetic Field from a Quarter Circle) Suppose there is a wire in the shape shown below with a steady current I travelling clockwise. What is the magnetic field at point P?

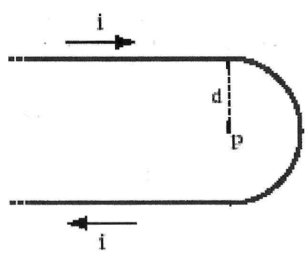


5. A long wire is bent into the hairpin-like shape shown in the figure.

   (a) What is the direction of the magnetic field at the indicated point P, which lies at the center of the half-circle?

   (b) What is the magnitude of the magnetic field at that point? (Hint: At the center of a circular current loop of radius R the loops magnetic field has magnitude $B_{\text{loop}} = \frac{\mu_0 I}{2R}$.)

   (c) Suppose an electron is at rest at point P. What force does the magnetic field exert on it?



6. Problem:

Problem 19.5: In the figure below, a long circular pipe with outside radius R carries a (uniformly distributed) current I into the page. A wire runs parallel to the pipe at a distance of $3R$ from center to center. Find the magnitude and direction (into or out of the page) of the current in the wire such that the net magnetic field at point P has the same magnitude as the net magnetic field at the center of the pipe but is in the opposite direction.